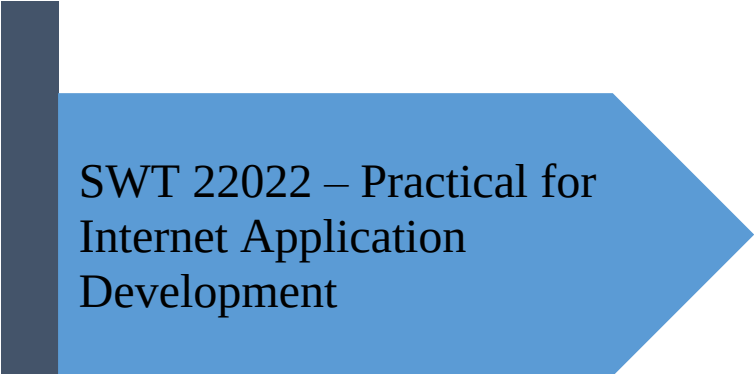




SWT 22022 – Practical for
Internet Application
Development

Department of Information
and Communication
Technology
Faculty of Technology



Lab sheet :17
Reg. Number: SEU/IS/20/ICT/084
Academic Year :2020/2021
Date: 2024.10.21
Practical No :17

Title: Introduction to React and Basic Functional Components

Aims:

- Setting up development environment
- Project creation
- Understanding the core concepts

Tasks:

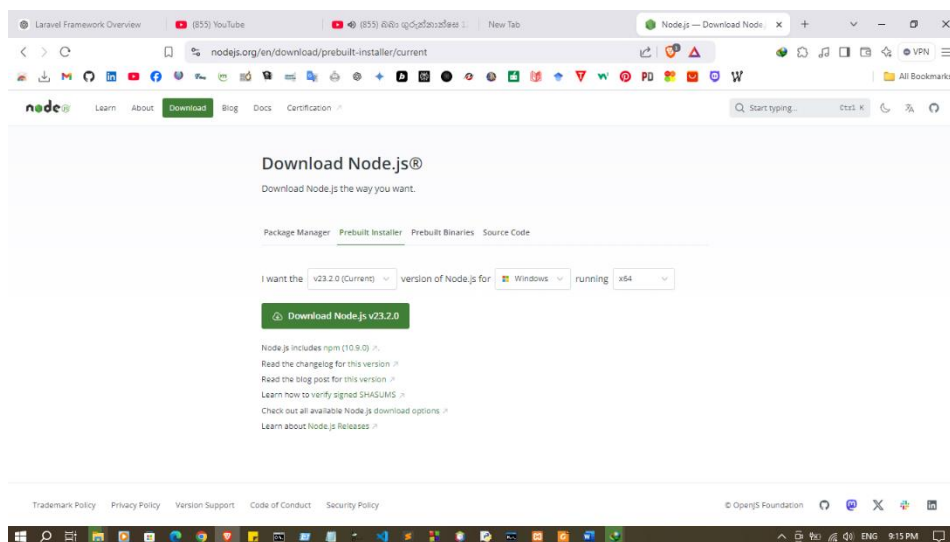
- Install Node.js and npm
- Setting up React
- Start the development server
- Basic React Components

Activities :

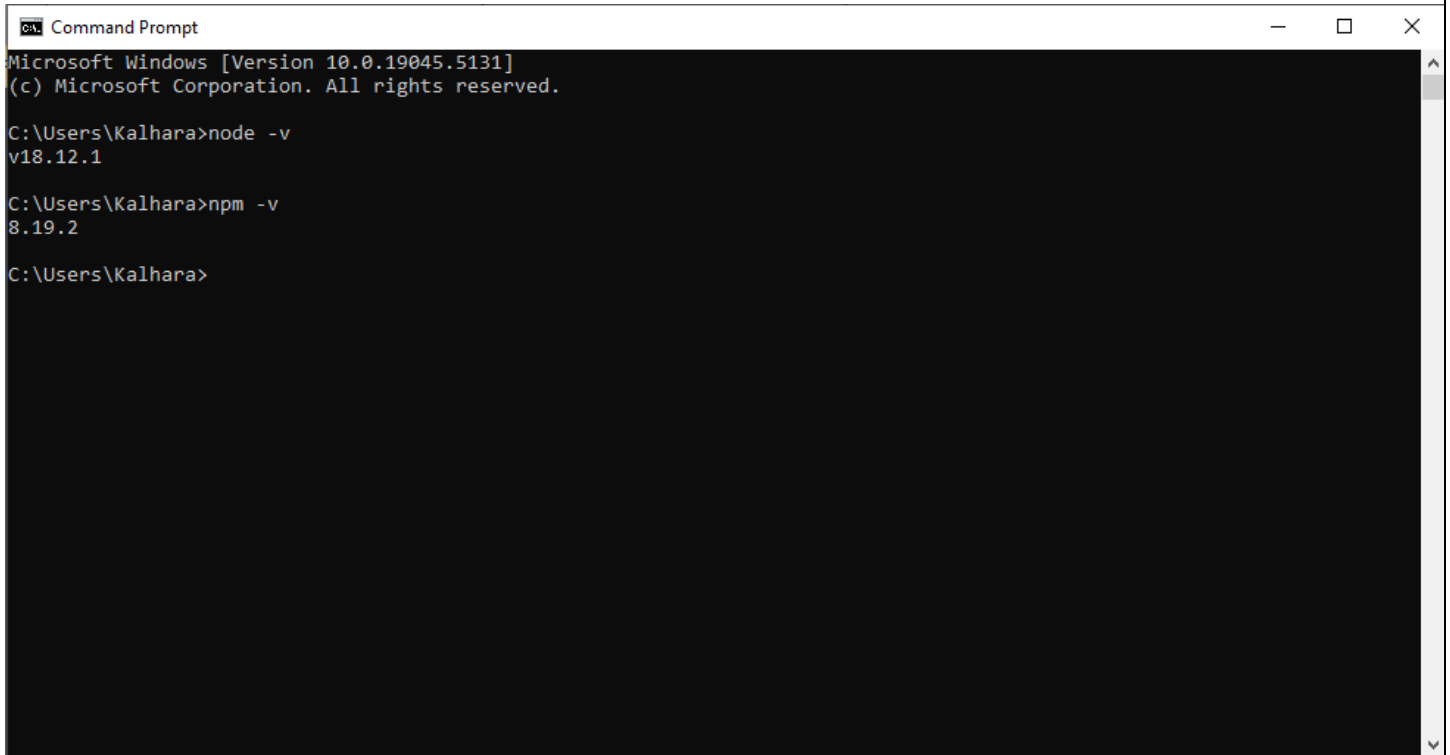
1. Install Node.js and npm

Windows:

- Download the appropriate Node.js installer from the official website:
<https://node.js.org/en/download/>
- Run the downloaded installer file and follow the on-screen instructions.



- Verify the installation by opening a command prompt and typing
node -v and npm -v



```
Command Prompt
Microsoft Windows [Version 10.0.19045.5131]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Kalhara>node -v
v18.12.1

C:\Users\Kalhara>npm -v
8.19.2

C:\Users\Kalhara>
```

macOS:

- Install Homebrew if you don't already have it: <https://docs.brew.sh/Installation>
- Once Homebrew is installed, run the following command to install Node.js:
brew install node
- Verify the installation by opening a Terminal window and typing
node -v and npm -v

Linux:

- The installation method for Linux depends on your specific distribution. However, generally, you can use your package manager to install Node.js. For example, on Ubuntu, you can use the following command:
sudo apt install nodejs npm
- Verify the installation by opening a terminal window and typing
node -v and npm -v

2. Setting up React

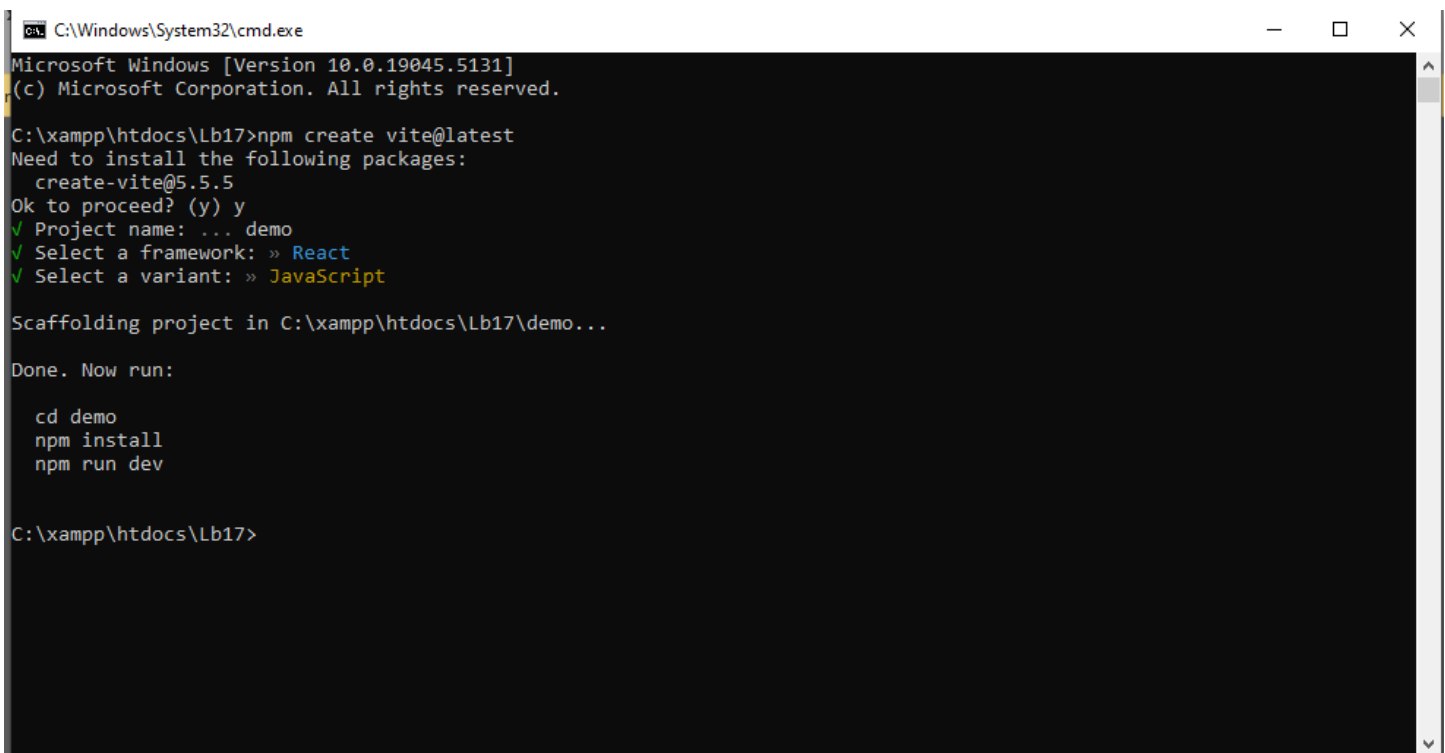
```
E:\SEUSL\AS-02\SWT22022\NEW\React>npm create vite@latest
✓ Project name: ... demo
✓ Select a framework: » React
✓ Select a variant: » JavaScript

Scaffolding project in E:\SEUSL\AS-02\SWT22022\NEW\React\demo...

Done. Now run:

  cd demo
  npm install
  npm run dev

E:\SEUSL\AS-02\SWT22022\NEW\React>
```



```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.5131]
(c) Microsoft Corporation. All rights reserved.

C:\xampp\htdocs\Lb17>npm create vite@latest
Need to install the following packages:
  create-vite@5.5.5
Ok to proceed? (y) y
✓ Project name: ... demo
✓ Select a framework: » React
✓ Select a variant: » JavaScript

Scaffolding project in C:\xampp\htdocs\Lb17\demo...

Done. Now run:

  cd demo
  npm install
  npm run dev

C:\xampp\htdocs\Lb17>
```

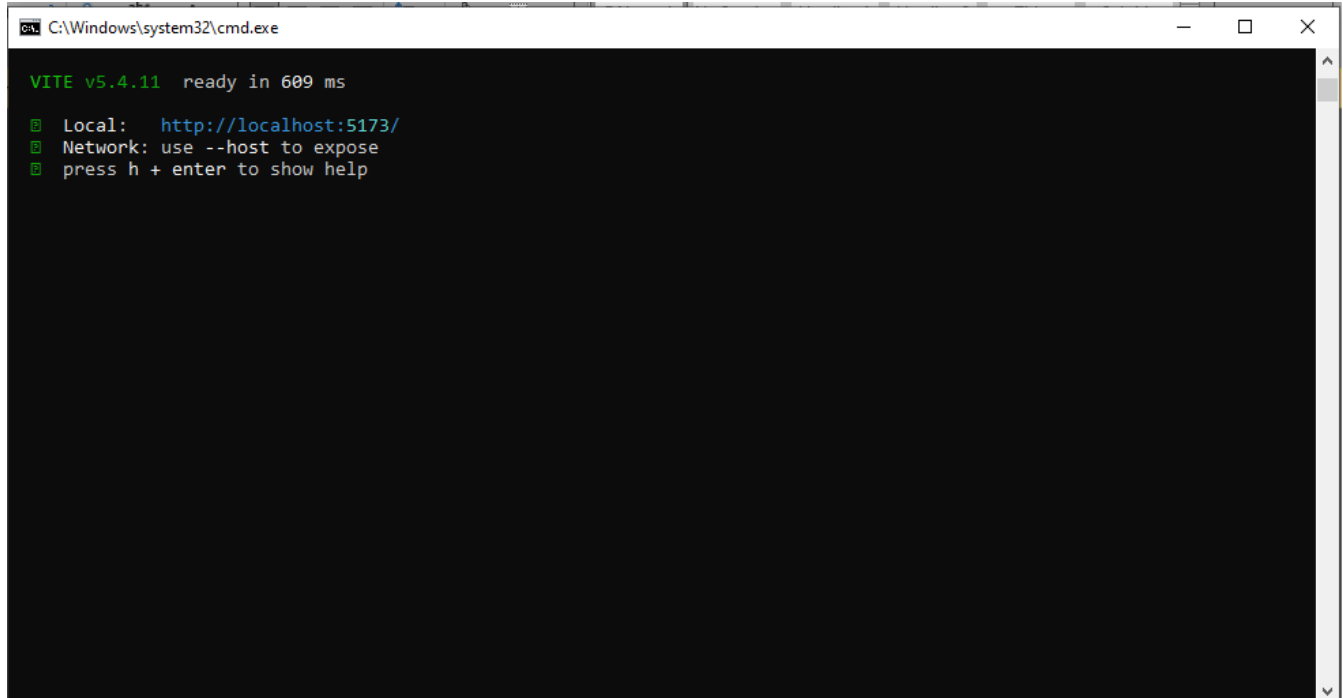
3. Start the development server

Install the required dependencies for the project:

npm install

Run the development server:

```
E:\SEUSL\AS-02\SWT22022\NEW\React\demo>npm run dev  
  
> demo@0.0.0 dev  
> vite  
  
VITE v5.4.8 ready in 1291 ms  
  
📍 Local:   http://localhost:5173/  
📍 Network: use --host to expose  
📍 press h + enter to show help
```



```
C:\Windows\system32\cmd.exe  
  
VITE v5.4.11 ready in 609 ms  
  
📍 Local:   http://localhost:5173/  
📍 Network: use --host to expose  
📍 press h + enter to show help
```

4. Basic React Components

Hello.jsx

```
function Hello() {  
  return (  
    <>  
    <div>  
      <h2>Hello World !</h2>  
    </div>  
  )  
}
```

```

    </>
  )
}
export default Hello

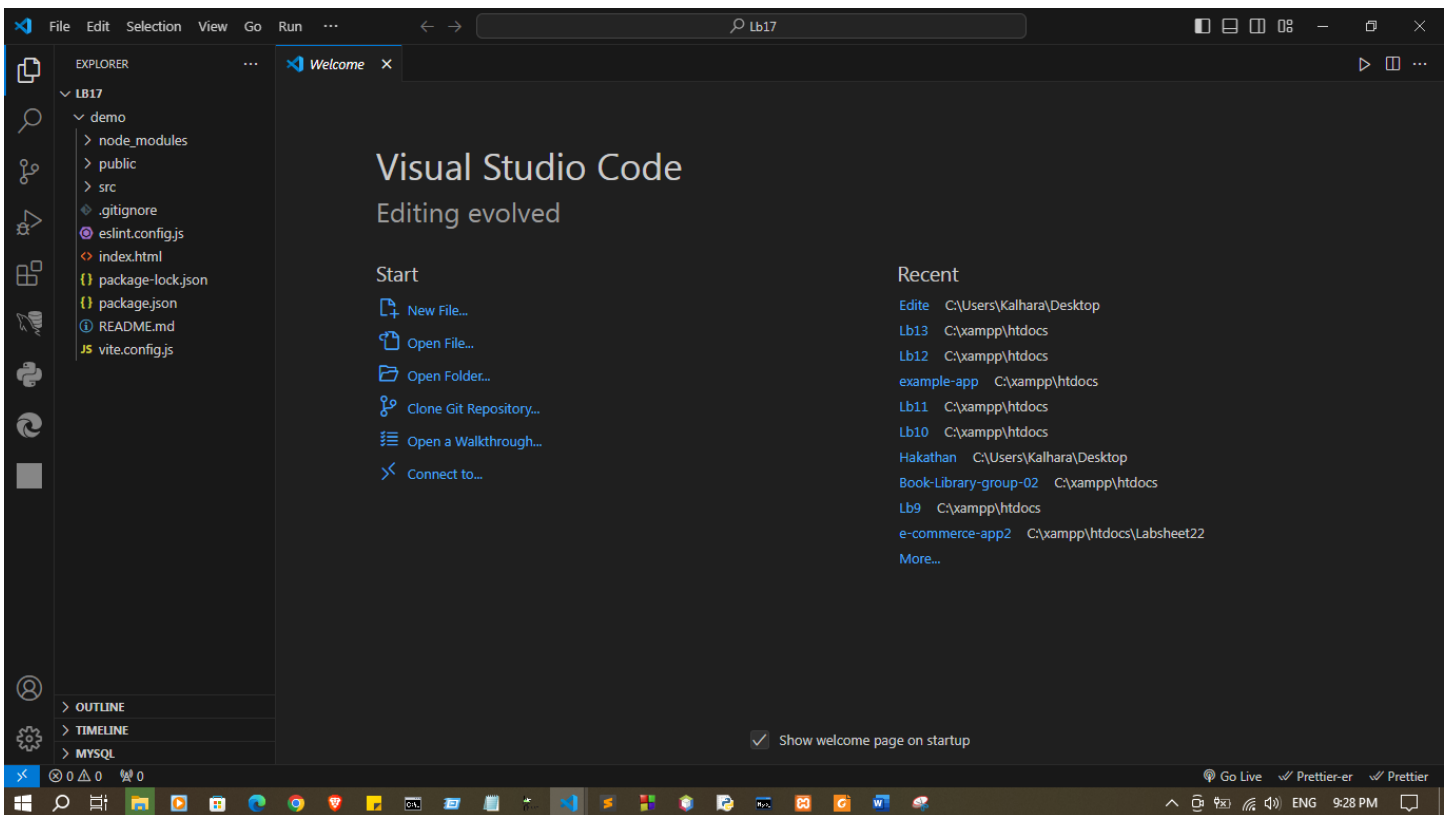
```

main.jsx

```

import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
// import App from './App.jsx'
import './index.css'
import Hello from './hello.jsx'
createRoot(document.getElementById('root')).render(
  <StrictMode>
    <Hello />
  </StrictMode>,
)

```

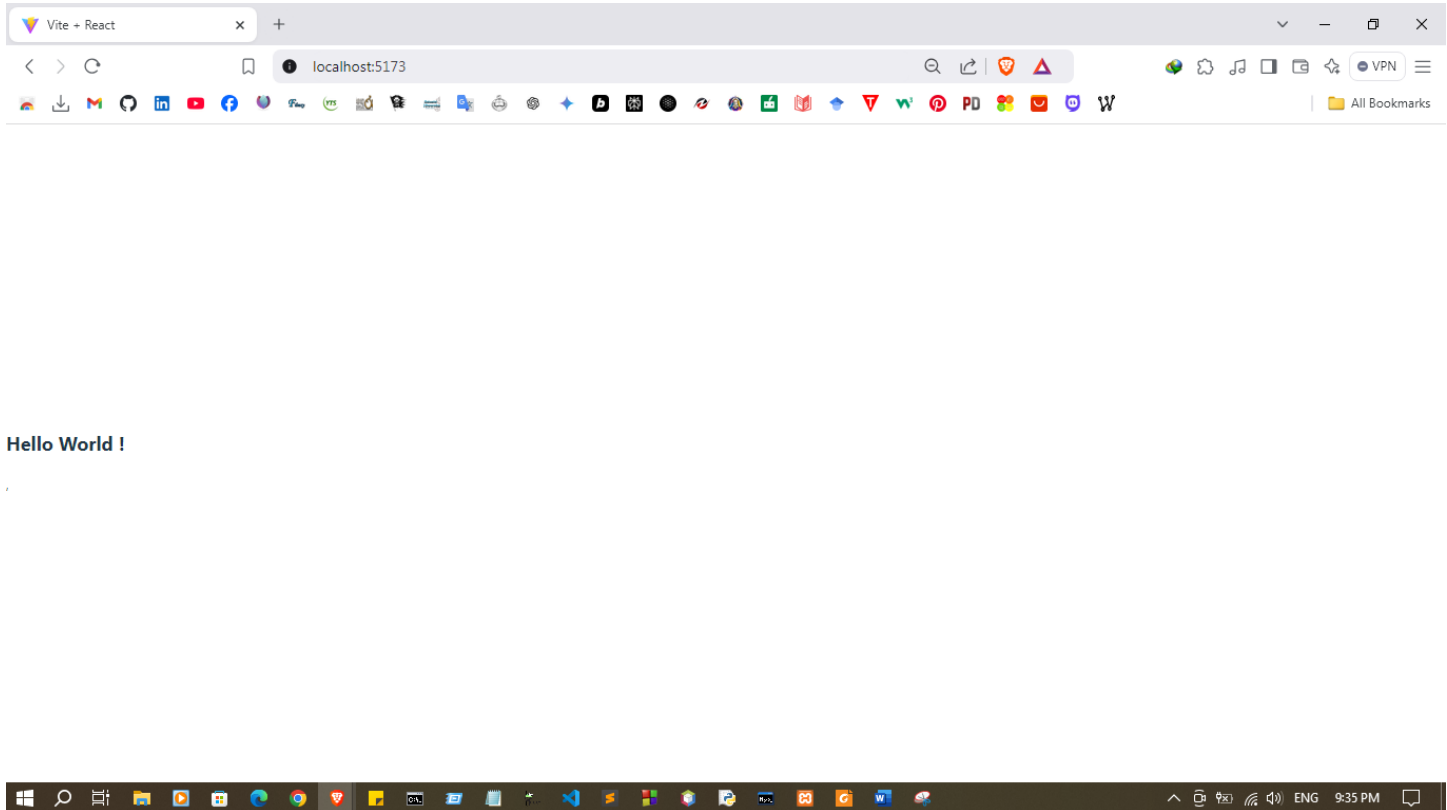


Hello.jsx

```
function Hello() {  
  return (  
    <>  
    <div>  
    <h2>Hello World !</h2>  
    </div>  
    </>  
  )  
}  
export default Hello
```

App.jsx

```
import { StrictMode } from 'react'  
import { createRoot } from 'react-dom/client'  
// import App from './App.jsx'  
import './index.css'  
import Hello from './Hello.jsx'  
function App() {  
  return (  
    <>  
    <StrictMode>  
    <Hello />  
    </StrictMode>,  
  
    </>  
  )  
}  
export default App
```



Discussion:

- This lab session introduces React, a popular JavaScript library for building user interfaces, with a focus on basic functional components. It starts with setting up the development environment by installing Node.js and npm, which are essential tools for running and managing React projects. Participants will then learn to set up a React application and start the development server to view real-time updates as they build. The session emphasizes understanding React's core concepts, such as components, props, and the virtual DOM, while creating and working with basic functional components. By the end of this session, students will have a foundational understanding of how to structure and develop a React project effectively.